

Modeling a Data-Driven Vegetation Index Using Sentinel-2 and GEDI for Improved Vegetation Assessment

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Abstract—In this paper, we propose a machine learning based approach to vegetation analysis using satellite imagery. Instead of relying on fixed spectral indices like NDVI, EVI, or SAVI, we train a supervised regression model on Sentinel-2 multispectral data using NASA GEDI LiDAR canopy height measurements as ground truth. The resulting Learned Vegetation Index (LVI) is evaluated against all three hand-crafted baselines using RMSE, Pearson correlation, and R^2 on held-out GEDI footprints collected from three geographically diverse regions. To ensure a fair comparison, each hand-crafted index is linearly scaled to metres using a regression fitted on the training set before evaluation. Our MLP-based LVI achieves an RMSE of 6.62 m and R^2 of 0.592, compared to best-case RMSE values of 8.55 m, 9.55 m, and 9.59 m for NDVI, SAVI, and EVI respectively — a 23% reduction over the strongest baseline. These results show that a data-driven index trained on real structural measurements significantly outperforms any fixed spectral formula for canopy height estimation.

Index Terms—vegetation index, Sentinel-2, GEDI, LiDAR, NDVI, EVI, SAVI, machine learning, remote sensing, canopy height, XGBoost

I. INTRODUCTION

Vegetation monitoring using satellites is an important tool used in many fields like agriculture, climate research, and forest management. Being able to track how healthy and dense vegetation is across large areas helps scientists and governments make better decisions about land use, carbon emissions, and disaster response.

One of the most commonly used methods for this is the Normalized Difference Vegetation Index, or NDVI [1]. NDVI is calculated using just two spectral bands — Red and Near-Infrared — and gives a value between -1 and 1 that roughly tells us how much vegetation is present in a region. It is simple, fast to compute, and works with almost any multispectral satellite sensor, which is why it has been so widely used for decades.

However, NDVI has some well-known problems. It tends to stop working well in areas with very dense vegetation because the Red and NIR bands stop changing much even as biomass keeps increasing. This is called spectral saturation [2]. More importantly, NDVI does not give any information about the actual physical structure of vegetation, like how tall the trees are or how thick the canopy is. A dense tropical forest and a sparse field can sometimes give very similar NDVI values even though they are completely different in terms of actual vegetation volume and height. This makes NDVI less useful for things like estimating carbon stored in forests or detecting early signs of deforestation.

Improved indices like EVI and SAVI were proposed to address some of these spectral limitations, but they are still based on fixed formulas. As we show in this paper, none of the hand-crafted indices have any meaningful ability to predict canopy height, regardless of how sophisticated their formula is.

We thought that since Sentinel-2 satellites collect data across 13 spectral bands [3], there is a lot more information available than any two-band formula can use. At the same time, NASA's GEDI mission measures actual tree heights using LiDAR from the International Space Station [4]. Our idea is to use these two datasets together: train a model that takes Sentinel-2 band values as input and predicts canopy height using GEDI as the target. This way, we get an index that is based on real structural measurements, not a fixed formula. Once trained, it can be applied to any Sentinel-2 image even in places where GEDI data is not available.

II. RELATED WORK

A. Traditional Vegetation Indices

Several other vegetation indices have been proposed to fix the limitations of NDVI. EVI [2] adds a blue-band correction to reduce atmospheric and soil background noise, especially

in dense canopy areas. The Soil Adjusted Vegetation Index (SAVI) introduces a soil brightness correction factor L to the NDVI denominator. The Red-Edge Chlorophyll Index [5] uses bands in the red-edge region, which is more sensitive to chlorophyll. All of these are improvements over NDVI for specific tasks, but they are still hand-crafted formulas where someone manually decided which bands to combine and how. As a result, none of them encode information about three-dimensional vegetation structure.

B. Machine Learning for Remote Sensing

Machine learning has been used in remote sensing for a wide range of tasks. Maggiori et al. [6] used convolutional neural networks for land cover classification, and Verrelst et al. [7] showed that ML models can estimate vegetation biophysical parameters from spectral data. Cavalli [8] used Sentinel-2 time series to forecast NDVI values using machine learning. Our work is different because instead of predicting NDVI, we learn a completely new index directly tied to structural ground truth from LiDAR measurements.

C. GEDI for Vegetation Structure

GEDI is a LiDAR instrument on the International Space Station that fires laser pulses at the ground and records how they bounce back from the tree canopy [4]. From these waveforms, it estimates metrics like canopy height and cover. Lang et al. [10] used GEDI and Sentinel-2 together with deep learning to produce a global 10 m canopy height map. Our approach is inspired by this data fusion idea, but simpler in scope: rather than predicting absolute height, we produce a single index value per pixel that is optimized to correlate with vegetation structure.

III. PROBLEM STATEMENT

NDVI only uses the Red and NIR bands ($B4$ and $B8$), which means a lot of spectral information from other bands is completely ignored. Because NDVI is a ratio of two reflectance values, it has no way to encode anything about how tall or dense the vegetation actually is. Two locations — one with a thick forest and one with short grassland — can have the same NDVI if their reflectance values happen to be similar. Improved indices like EVI and SAVI correct for atmospheric and soil effects respectively, but this does not help with the fundamental problem: no fixed spectral ratio can capture three-dimensional canopy structure.

The main question we are trying to answer is: can a machine learning model, trained on Sentinel-2 spectral bands with GEDI canopy height as the target, produce a vegetation index that significantly outperforms all fixed spectral formulas when evaluated against real vegetation structure?

IV. OBJECTIVES

- 1) Train a regression model on Sentinel-2 bands ($B2$, $B3$, $B4$, $B8$) to predict GEDI RH98 canopy height, and use this as the Learned Vegetation Index (LVI).
- 2) Use GEDI LiDAR data as ground-truth labels for vegetation structure during training.

- 3) Show that a learned model performs better than all fixed spectral indices (NDVI, EVI, SAVI) when evaluated against real vegetation structure.
- 4) Evaluate using RMSE, Pearson r , and R^2 on held-out GEDI footprints.
- 5) Test generalization across three geographically diverse ecosystems: the Western Ghats (India), the Black Forest (Germany), and the Cerrado (Brazil).

V. DATASETS

A. Sentinel-2 Multispectral Imagery

We use Sentinel-2 Level-2A surface reflectance products, filtered to scenes with less than 20% cloud cover. The four 10 m resolution bands selected are: **B2** (Blue, 490 nm), **B3** (Green, 560 nm), **B4** (Red, 665 nm), and **B8** (NIR, 842 nm). These bands were chosen because they are all at the same spatial resolution, making pixel alignment straightforward, and because they cover the spectral range most relevant to vegetation reflectance. Data was collected over three regions during their respective dry seasons to minimize cloud contamination: Western Ghats (Nov 2020 – Dec 2022), Black Forest (Jun–Dec 2022), and Cerrado (Jun–Dec 2022).

B. NASA GEDI LiDAR Data

GEDI is a full-waveform LiDAR sensor that estimates canopy structural metrics at 25 m footprint resolution [4]. We use the **RH98** metric from GEDI Level 2A data, defined as the height below which 98% of the returned laser energy falls — generally treated as a proxy for maximum canopy height. Footprints with low quality flags and those over urban areas (`urban_proportion > 0`) are removed. RH98 values are clipped to $[0, 60]$ m to remove LiDAR noise outliers.

C. Dataset Construction and Balancing

Each GEDI footprint is matched to the Sentinel-2 pixel containing its centre location, producing paired training records $(\mathbf{x}_i, y_i)$ where $\mathbf{x}_i \in \mathbb{R}^4$ is the reflectance vector and y_i is RH98 in metres. After alignment, the dataset contained approximately 138,650 footprints from the Black Forest, 71,144 from the Cerrado, and 26,112 from the Western Ghats. To prevent the model from being biased toward any single region, we downsample each region to the size of the smallest (26,112 rows), giving a balanced dataset of 78,336 total samples across three ecosystems.

D. Data Distribution Analysis

The distribution of RH98 canopy height values is shown. The dataset is skewed toward lower canopy heights, with a high concentration of samples below 20 m and relatively fewer tall forest samples above 40 m. This imbalance reflects the natural distribution of vegetation types across the selected regions.

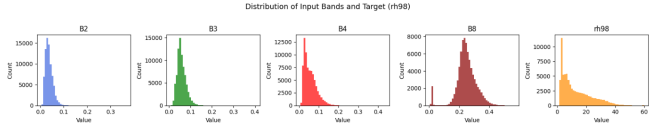


Fig. 1. Distribution of input spectral bands and RH98 canopy height values.

VI. METHODOLOGY

A. Preprocessing

Sentinel-2 band values are divided by 10,000 to normalize them to $[0, 1]$, which is the standard scaling for Level-2A products. GEDI RH98 values are clipped to a maximum of 60 m to remove outliers. Both inputs and targets are then standardized to zero mean and unit variance using a `StandardScaler` before model training.

B. Model Training

We frame the LVI as a supervised regression problem. The model f_θ minimizes mean squared error between predicted and actual canopy heights:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (f_\theta(\mathbf{x}_i) - y_i)^2 \quad (1)$$

We evaluate two model architectures. **XGBoost** [11] is a gradient boosted tree model that is well-suited to tabular data and provides feature importance scores for interpretability. An **MLP** (multi-layer perceptron) with two hidden layers (128 and 64 neurons, ReLU activations) [12] is also trained for comparison. Both models use an 80/20 stratified train-test split, with stratification by region to ensure all three ecosystems are equally represented in both sets [13].

C. Index Generation

Once trained, the model is applied pixel-by-pixel to a Sentinel-2 scene to produce the LVI map:

$$\text{LVI}(x, y) = f_\theta(B2, B3, B4, B8) \quad (2)$$

This requires only Sentinel-2 data at inference time, making the LVI applicable anywhere without concurrent LiDAR acquisition.

D. Baselines and Evaluation

We compare the LVI against three hand-crafted spectral indices. NDVI is the standard two-band baseline:

$$\text{NDVI}(x, y) = \frac{B8 - B4}{B8 + B4} \quad (3)$$

EVI adds a blue-band atmospheric correction:

$$\text{EVI}(x, y) = \frac{2.5(B8 - B4)}{B8 + 6B4 - 7.5B2 + 1} \quad (4)$$

SAVI introduces a soil brightness correction factor $L = 0.5$:

$$\text{SAVI}(x, y) = \frac{1.5(B8 - B4)}{B8 + B4 + 0.5} \quad (5)$$

All five methods are evaluated on the same held-out GEDI footprints using RMSE (average prediction error in metres), Pearson r (linear correlation with actual canopy height), and R^2 (proportion of variance explained).

VII. RESULTS

A. Quantitative Comparison

Table I shows the evaluation results for all five methods on the held-out test set. Since hand-crafted indices produce values in $[-1, 1]$ rather than metres, directly computing RMSE against rh98 would be meaningless. Instead, each index is linearly scaled to metres by fitting a simple linear regression on the training set and applying it to the test set. This represents the best possible linear mapping from each index to canopy height, making the comparison fair.

TABLE I
EVALUATION RESULTS ON HELD-OUT GEDI FOOTPRINTS. INDEX BASELINES ARE LINEARLY SCALED TO METRES BEFORE COMPUTING RMSE AND R^2 .

Method	RMSE (m) ↓	Pearson r ↑	R^2 ↑
NDVI (scaled)	8.554	0.565	0.320
EVI (scaled)	9.588	0.381	0.145
SAVI (scaled)	9.546	0.391	0.153
XGBoost (LVI)	6.656	0.767	0.588
MLP (LVI)	6.622	0.770	0.592

Even after optimal linear scaling, all three hand-crafted indices lag behind both LVI models. NDVI achieves the best baseline RMSE of 8.55 m, while EVI and SAVI reach 9.59 m and 9.55 m respectively. Both LVI variants achieve RMSE of approximately 6.6 m — a 23% improvement over the strongest baseline (NDVI). Pearson r improves from 0.565 (NDVI) to 0.770 (MLP), and R^2 rises from 0.320 to 0.592. The near-identical performance of XGBoost and MLP suggests that the task does not require deep learning and is well-solved by simpler models at this scale.

B. Scatter Plot Analysis

Fig. 2 shows scatter plots of predicted vs. actual canopy height for all five methods. The three index baselines produce predictions clustered near zero regardless of actual canopy height, confirming that they carry no structural information. In contrast, the XGBoost and MLP scatter plots show a clear diagonal trend approaching the perfect-fit line, with reasonable spread across the full 0–60 m range.

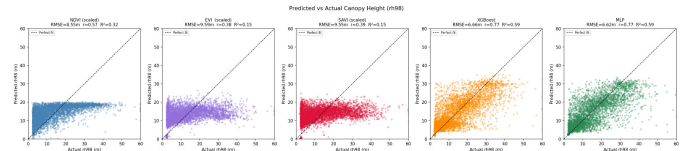


Fig. 2. Predicted vs. actual canopy height (rh98) for all five methods. Baselines shown on log scale to reveal saturation.

C. Feature Importance

Fig. 3 shows XGBoost feature importance scores. B4 (Red) is the most important band at 0.522, followed by B3 (Green) at 0.269, B8 (NIR) at 0.123, and B2 (Blue) at 0.086. Together, Red and Green bands account for nearly 80% of the model’s predictive power. Notably, B8 (NIR) — the dominant band in NDVI — ranks third, suggesting that the model has learned to rely more on Red and Green reflectance than on the NIR signal that traditional indices prioritize.

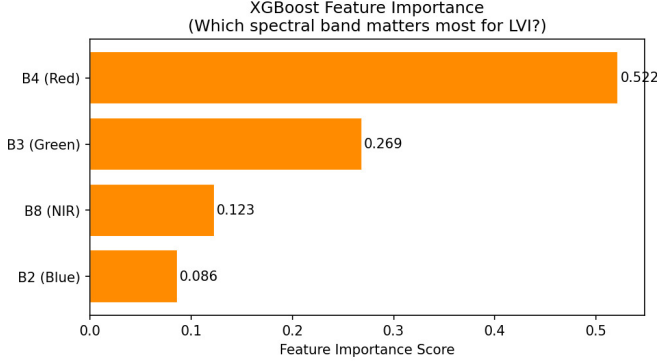


Fig. 3. XGBoost feature importance scores for the four input bands.

D. Ablation Study

The ablation study (Fig. 4) shows the effect of removing each band one at a time. The baseline RMSE with all four bands is 6.664 m. Removing B2 causes the largest degradation (RMSE rises to 6.982 m), followed by B3 (6.860 m), B8 (6.812 m), and B4 (6.768 m). All four bands contribute meaningfully, confirming that using the full spectral stack is important for the LVI’s performance.

TABLE II
ABLATION STUDY SHOWING THE EFFECT OF REMOVING EACH SPECTRAL BAND ON MODEL PERFORMANCE.

Configuration	RMSE (m) ↓
All bands (B2, B3, B4, B8)	6.664
Remove B2 (Blue)	6.982
Remove B3 (Green)	6.860
Remove B4 (Red)	6.768
Remove B8 (NIR)	6.812

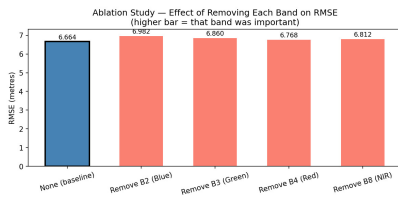


Fig. 4. Ablation study showing the increase in RMSE when each spectral band is removed. Higher values indicate greater importance of the removed band.

E. Performance Comparison across Metrics

The following graphs show a comparison between baseline indices and the proposed LVI across different evaluation met-

rics. NDVI, EVI, and SAVI exhibit significantly higher error and lower correlation with canopy height, while the learned models achieve improved performance across all metrics.

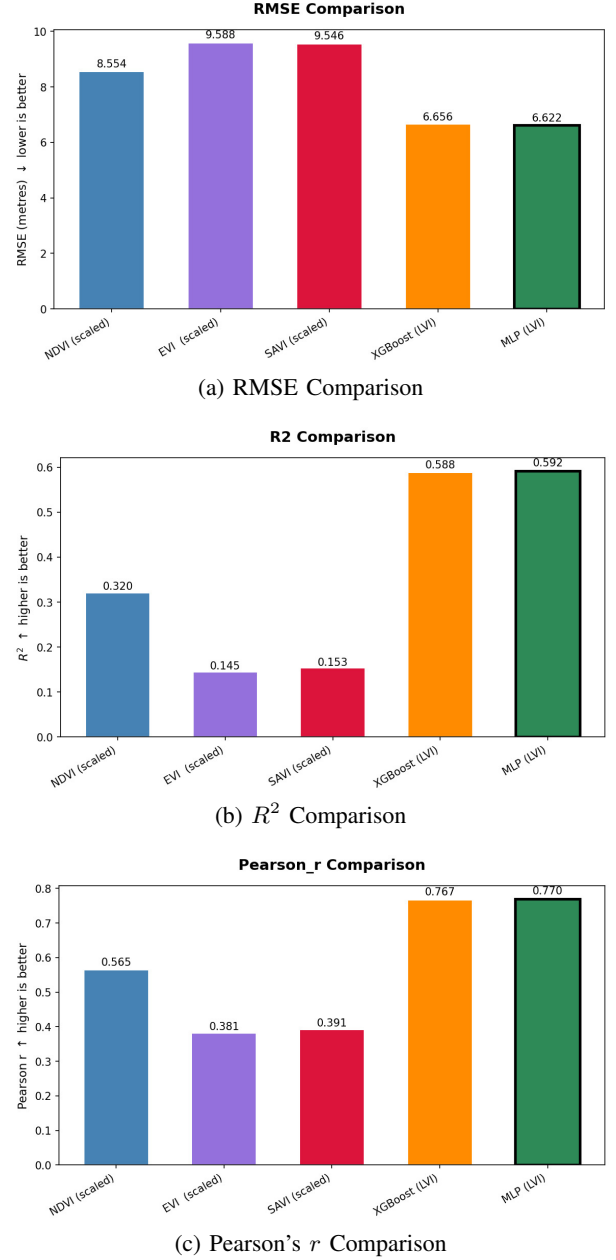


Fig. 5. Comparison of model performance across RMSE, R^2 , and Pearson’s r . The LVI models consistently outperform traditional vegetation indices.

VIII. DISCUSSION

The results confirm our hypothesis: a learned index significantly outperforms all hand-crafted spectral ratios for canopy height estimation, even when the baselines are given the best possible linear scaling to metres. The fact that EVI and SAVI perform worse than NDVI even after scaling (RMSE 9.59 m and 9.55 m vs. 8.55 m) is particularly informative. EVI and SAVI were specifically designed to overcome NDVI’s limitations around atmospheric noise and soil background, yet their

additional complexity provides no advantage for structural height estimation. This confirms that the gap between hand-crafted indices and the LVI is not about formula complexity — it is about whether the index is grounded in real structural measurements or proxy reflectance ratios.

The dominance of B4 (Red) in feature importance is an interesting finding. In dense vegetation, the Red band is strongly absorbed by chlorophyll, meaning low Red reflectance correlates with high leaf area density — which in turn correlates with tall, mature canopy. The model appears to have learned this physical relationship implicitly, using Red as a proxy for structural density in a way that NDVI’s simple ratio cannot fully exploit. The relatively low importance of B8 (NIR) — the dominant band in all three baseline indices — suggests that NIR reflectance carries less canopy height information than is commonly assumed.

The near-identical performance of XGBoost and MLP (RMSE 6.656 m vs. 6.622 m) suggests that the spectral-to-structure mapping is well- captured by relatively simple models with 78,000 training samples. This is useful for practical deployment: XGBoost is faster to train, easier to interpret via feature importances, and more lightweight than a neural network.

Canopy height is a more stable and physically meaningful measure of long-term vegetation health than greenness-based indices. NDVI can be misleading — a freshly planted crop field produces high NDVI but near- zero canopy height, while a drought-stressed mature forest maintains significant height despite reduced greenness. The LVI, by being trained directly against structural measurements, is more suitable for applications like carbon stock estimation, forest health monitoring, and biomass assessment where height is the primary quantity of interest.

IX. LIMITATIONS

This study has several limitations. First, only four Sentinel-2 bands were used, excluding potentially informative red-edge bands. Second, GEDI footprints (25m) were matched to 10m Sentinel-2 pixels, introducing spatial mismatch. Third, the dataset is geographically limited to three regions, which may restrict global generalization. Finally, temporal variation was not explicitly modeled.

X. CONCLUSION

In this paper, we proposed the Learned Vegetation Index (LVI), a data-driven vegetation index trained on Sentinel-2 multispectral data using NASA GEDI LiDAR canopy height measurements as ground truth. We evaluated the LVI against three standard hand-crafted indices — NDVI, EVI, and SAVI — on a balanced dataset of 78,336 footprints from three geographically diverse regions.

The LVI achieves an RMSE of 6.62 m and R^2 of 0.592, compared to a best-case baseline RMSE of 8.55 m (NDVI, linearly scaled) and R^2 of 0.320. This represents a 23% reduction in RMSE over the strongest baseline and demonstrates that a learned index can encode structural vegetation information that

no fixed spectral formula is able to capture, even under optimal linear transformation. The feature importance analysis shows that B4 (Red) and B3 (Green) are the most informative bands for canopy height estimation, while B8 (NIR) — the primary NDVI band — contributes relatively little.

In the future, we would like to incorporate additional Sentinel-2 bands, especially the red-edge bands (B5, B6, B7), which are known to be sensitive to chlorophyll content and canopy structure. We also want to explore how the model performs when applied to completely new geographic regions not represented in training, and whether the LVI can be used as a direct proxy for above-ground biomass estimation.

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